

Operation Data Fortress

Digital Forensics & Incident Response Portfolio Report

Role: Forensic Lead

Scenario: ShadowGate Incident - Global Synergy Solutions (GSS)

This portfolio edition summarizes a simulated senior cybersecurity capstone investigation. It focuses on the technical investigation, evidence-handling approach, findings, and recommendations. Course-administrative material and other students' identifying information have been omitted.

Portfolio Focus

- Windows endpoint and digital forensic analysis
- Network traffic analysis with Wireshark
- AWS CloudTrail and Microsoft Azure Activity Log analysis
- Identity and access management review
- Evidence preservation and chain-of-custody practices
- Cross-source correlation and attack reconstruction

1. Executive Summary

Operation Data Fortress examined the simulated ShadowGate cybersecurity incident affecting Global Synergy Solutions (GSS). The investigation correlated endpoint, network, cloud, and identity evidence to identify the initial compromise, reconstruct attacker activity, assess impact, and preserve digital evidence using accepted forensic principles.

Evidence included Windows endpoint artifacts, Windows Security Event Logs, process information, registry modifications, network packet captures, AWS CloudTrail logs, Azure Activity Logs, IAM configuration data, and authentication records. The evidence supported a sequence beginning with a phishing-related compromise of workstation GSS-WS-101 and progressing into AWS and Microsoft Azure through compromised credentials.

The investigation identified persistence, credential theft, privilege abuse, unauthorized cloud administration, and access that placed sensitive organizational information at risk. The strongest conclusions came from correlating independent evidence sources rather than relying on a single artifact.

2. Scope and Objectives

- Identify the initial point of compromise and reconstruct the attack sequence.
- Analyze endpoint, network, cloud, authentication, and identity evidence.
- Identify malware execution, credential theft, privilege abuse, persistence, lateral movement, and unauthorized cloud activity.
- Preserve evidence integrity and maintain accountable handling.
- Produce technically supported findings and remediation recommendations.

3. Evidence Sources

Area	Evidence Reviewed	Investigative Purpose
Endpoint	Windows logs, processes, registry changes, autoruns	Execution, persistence, credential activity, system changes
Network	Packet capture / Wireshark	Suspicious communications and event correlation
AWS	CloudTrail, IAM, credential data	Authentication, enumeration, privilege and resource activity
Azure	Activity Logs, RBAC, identity records	Administrative actions, role changes, cloud persistence

4. Forensic Methodology

The investigation followed a structured workflow emphasizing preservation, examination, analysis, correlation, and documentation. Potentially volatile information was prioritized before detailed analysis, and original evidence was preserved while working copies were used whenever possible.

Endpoint Analysis

Workstation GSS-WS-101 was examined using Windows Event Viewer, process analysis, registry comparison, and Autoruns. Investigators reviewed security events, process activity, registry modifications, scheduled tasks, and autorun entries for evidence of malware execution, persistence, credential-related activity, and unauthorized system changes.

Network Analysis

Wireshark was used to examine packet-capture evidence for suspicious outbound communications, authentication activity, protocol usage, and possible command-and-control behavior. Network observations were correlated with endpoint activity and later cloud authentication events.

Cloud Analysis

AWS CloudTrail, Azure Activity Logs, IAM policies, RBAC assignments, credential information, and identity records were reviewed for unauthorized authentication, resource enumeration, administrative actions, privilege assignments, identity creation, and resource modifications.

Evidence Integrity

Evidence handling emphasized documentation, restricted access, secure storage, and integrity verification. Where applicable, cryptographic hash values were used to validate evidence, and transfers were documented through the chain-of-custody process.

5. Key Investigative Findings

Initial Endpoint Compromise

Evidence supported an initial compromise of GSS-WS-101 following execution of a malicious payload delivered through phishing. Abnormal process activity and system modifications appeared before unauthorized cloud activity. Registry and configuration changes indicated persistence intended to survive reboots.

Credential Theft and Privilege Abuse

Endpoint evidence indicated activity consistent with credential theft. Subsequent cloud authentication showed compromised credentials being used against AWS and Azure. Cloud activity included resource enumeration, role assumption, creation of privileged identities, and permission changes.

Multi-Cloud Activity and Persistence

AWS and Azure records documented administrative operations following compromised authentication. Endpoint persistence mechanisms and additional privileged cloud identities increased the complexity of eradication.

6. Correlation and Impact

Endpoint artifacts established the initial compromise, network captures documented communications associated with malicious activity, and cloud logs confirmed unauthorized authentication and administrative actions. The alignment of these independent sources increased confidence in the reconstructed sequence.

The incident affected both endpoint and cloud infrastructure. Evidence confirmed unauthorized access involving cloud identity services, administrative resources, and storage environments containing sensitive organizational information. Sensitive healthcare and financial information was placed at significant risk.

7. Reconstructed Attack Timeline

Stage	Reconstruction
1. Initial access	Phishing leads to malicious payload execution on GSS-WS-101.
2. Endpoint compromise	Abnormal process activity and system changes establish a foothold.
3. Persistence	Registry, autorun, and scheduled mechanisms support continued access.
4. Credential access	Credential-related activity precedes unauthorized cloud authentication.
5. Cloud access	Compromised credentials are used against AWS and Azure.
6. Privilege expansion	Role assumption, identity creation, permission changes, and resource enumeration occur.
7. Exposure risk	Unauthorized cloud access places sensitive organizational information at risk.

8. Analyst Challenges

The investigation required correlating technologies with different logging formats and distinguishing legitimate administrative activity from malicious actions performed with valid compromised credentials. Timing, sequence, and behavior were therefore evaluated alongside authentication success. Cross-source validation reduced reliance on isolated events.

9. Security Recommendations

- **Centralized logging:** Improve collection and correlation of endpoint, network, cloud, and identity telemetry.
- **Log retention:** Maintain sufficient retention to support retrospective investigations.
- **Endpoint detection and response:** Improve visibility into process execution, persistence, credential access, and system modification.
- **Identity controls:** Review privileged identities, role assignments, credential exposure, and unnecessary administrative permissions.
- **Forensic readiness:** Maintain standardized evidence-collection, preservation, and documentation procedures.
- **Exercises and training:** Conduct regular incident-response and forensic exercises.

10. Chain-of-Custody Approach

The project maintained chain-of-custody documentation for digital evidence throughout collection, storage, review, and transfer. Original evidence was kept unmodified where possible, analysis was performed on working copies, access was restricted to authorized investigators, and integrity checks were used where applicable.

11. Skills Demonstrated

- Digital forensics and incident-response methodology
- Windows endpoint artifact analysis
- Network forensics with Wireshark
- AWS CloudTrail and Azure Activity Log analysis
- IAM / RBAC and cloud identity analysis
- Cross-source event correlation and attack reconstruction
- Evidence preservation and chain-of-custody documentation
- Technical reporting and remediation recommendations

12. Conclusion

The ShadowGate investigation demonstrated how endpoint, network, cloud, and identity evidence can be correlated to reconstruct a complex incident. The evidence supported a progression from phishing and endpoint compromise through credential theft, persistence, unauthorized cloud access, privilege abuse, and exposure of sensitive resources. The project reinforced the importance of evidence integrity, multi-source validation, clear documentation, and forensic readiness.

References

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MITRE. (2023). *MITRE ATT&CK; for Enterprise*.

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National Institute of Standards and Technology. (2020). *Security and Privacy Controls for Information Systems and Organizations (NIST SP 800-53 Rev. 5)*.

U.S. Department of Health and Human Services. *Breach Notification Rule*.

Portfolio note: This is a condensed, sanitized presentation of a simulated academic cybersecurity investigation. It demonstrates investigative methodology and technical communication; it does not describe a production incident involving a real organization.